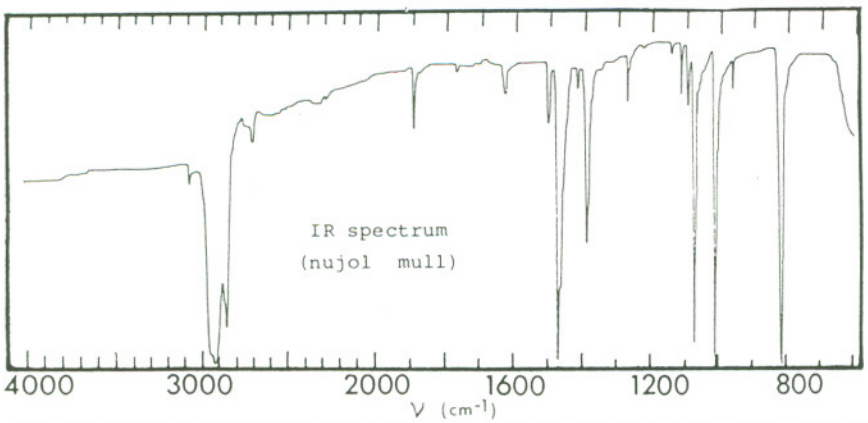
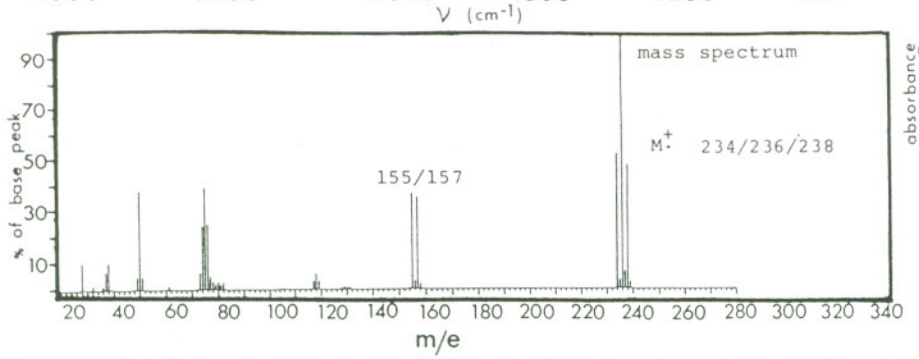
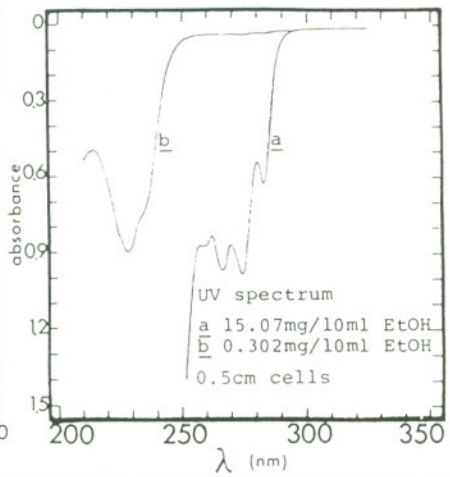
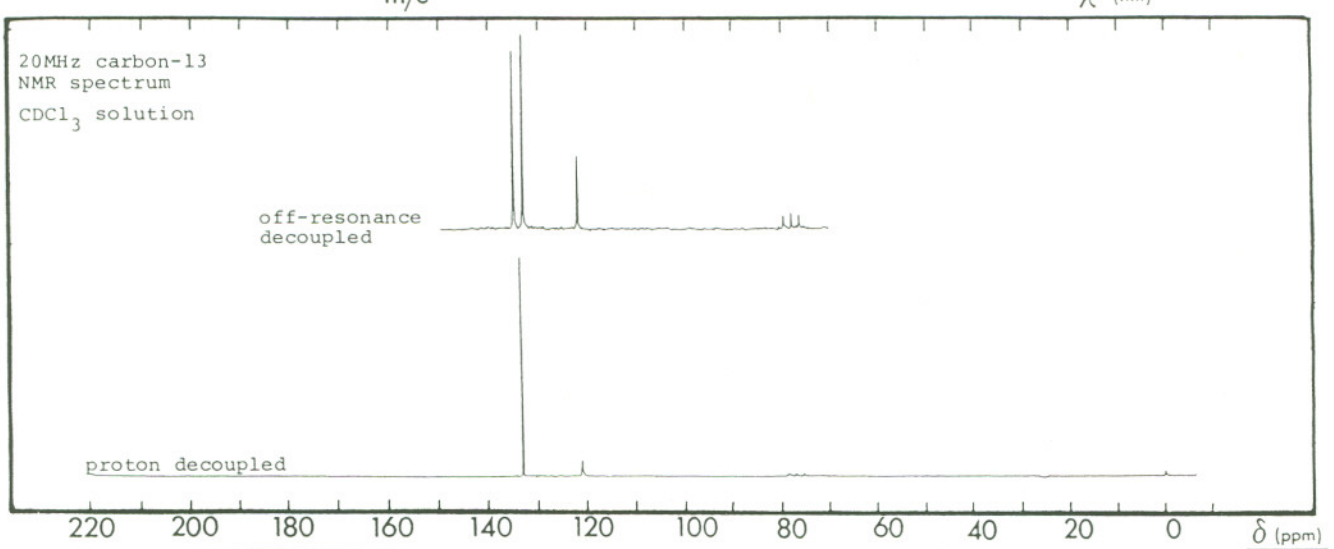
Organic Cume
November 14, 1992
Solve the structures for each set of spectra given below

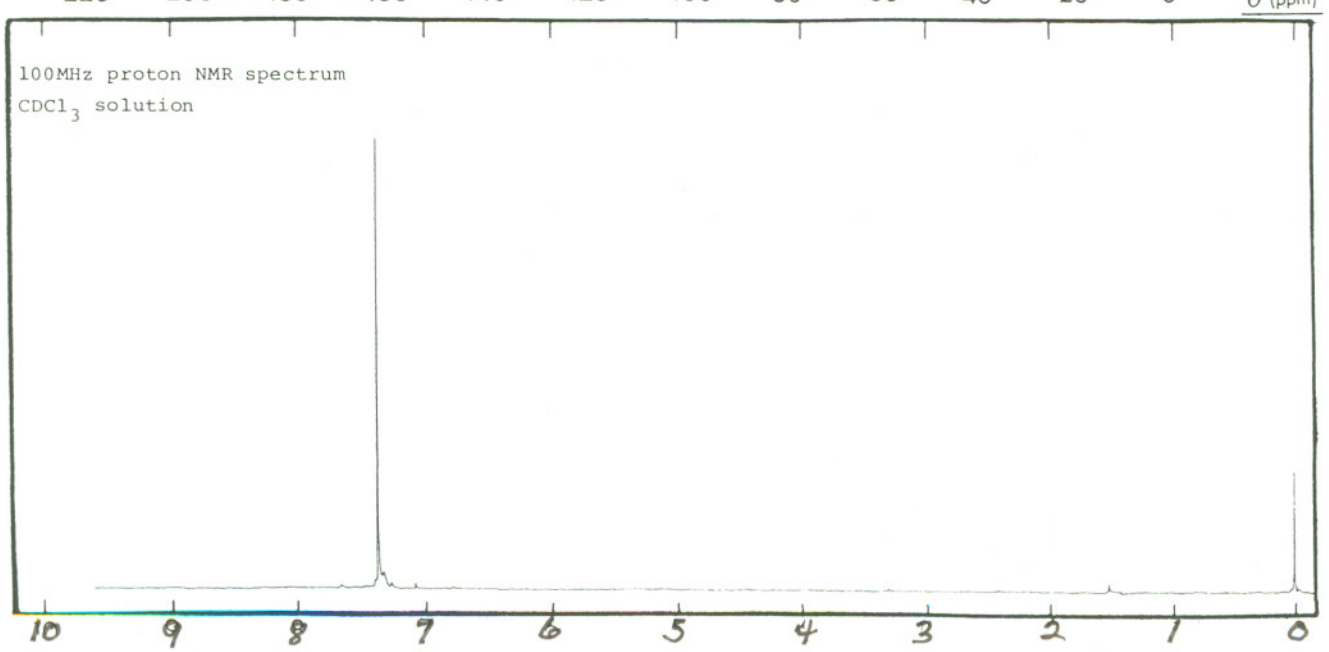
Problem 1



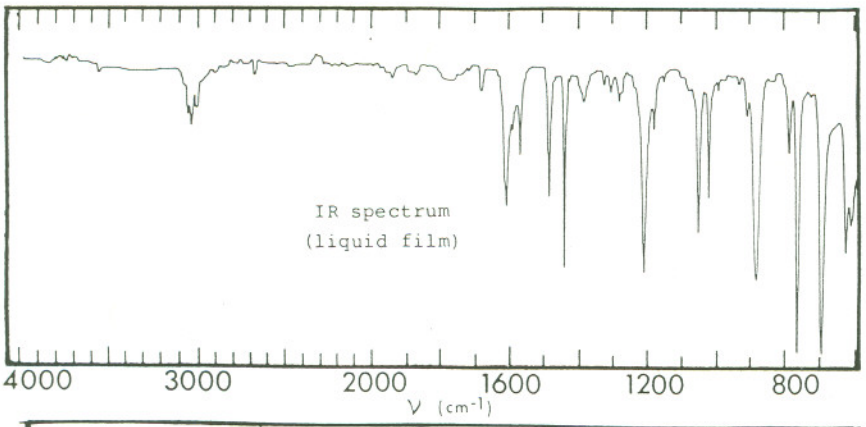


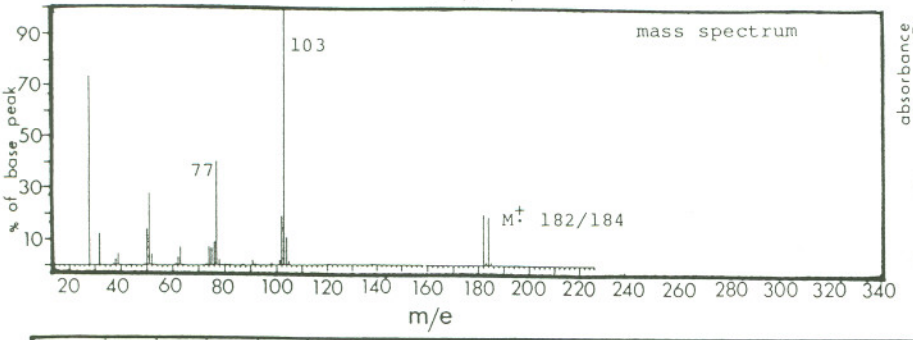


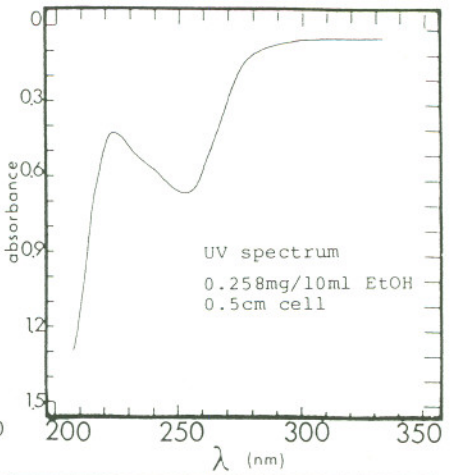


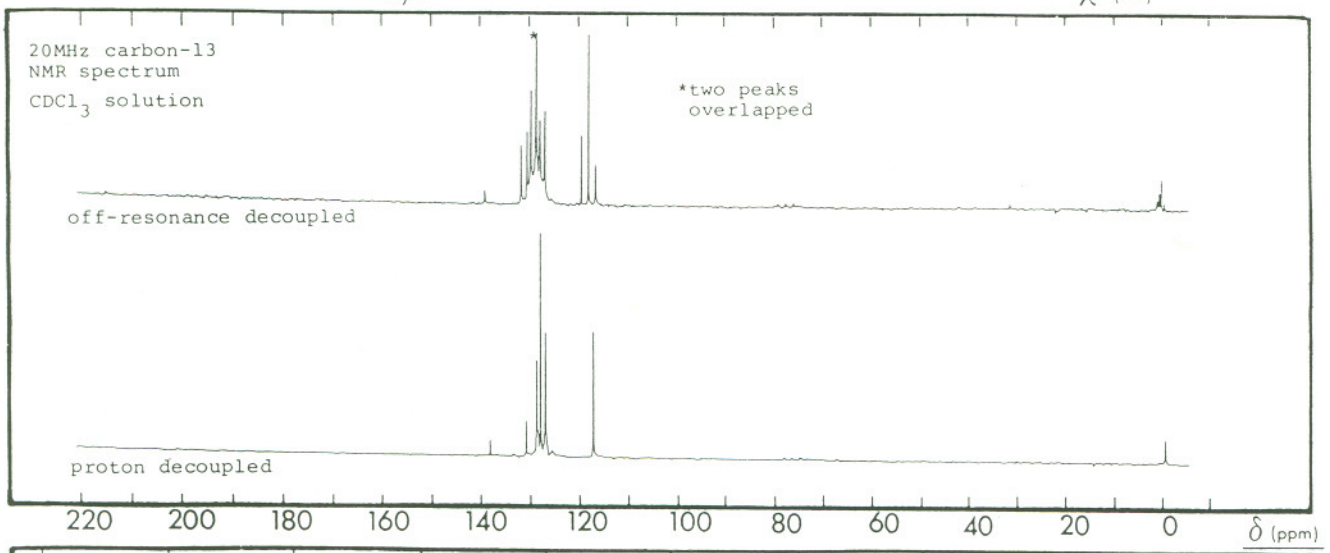


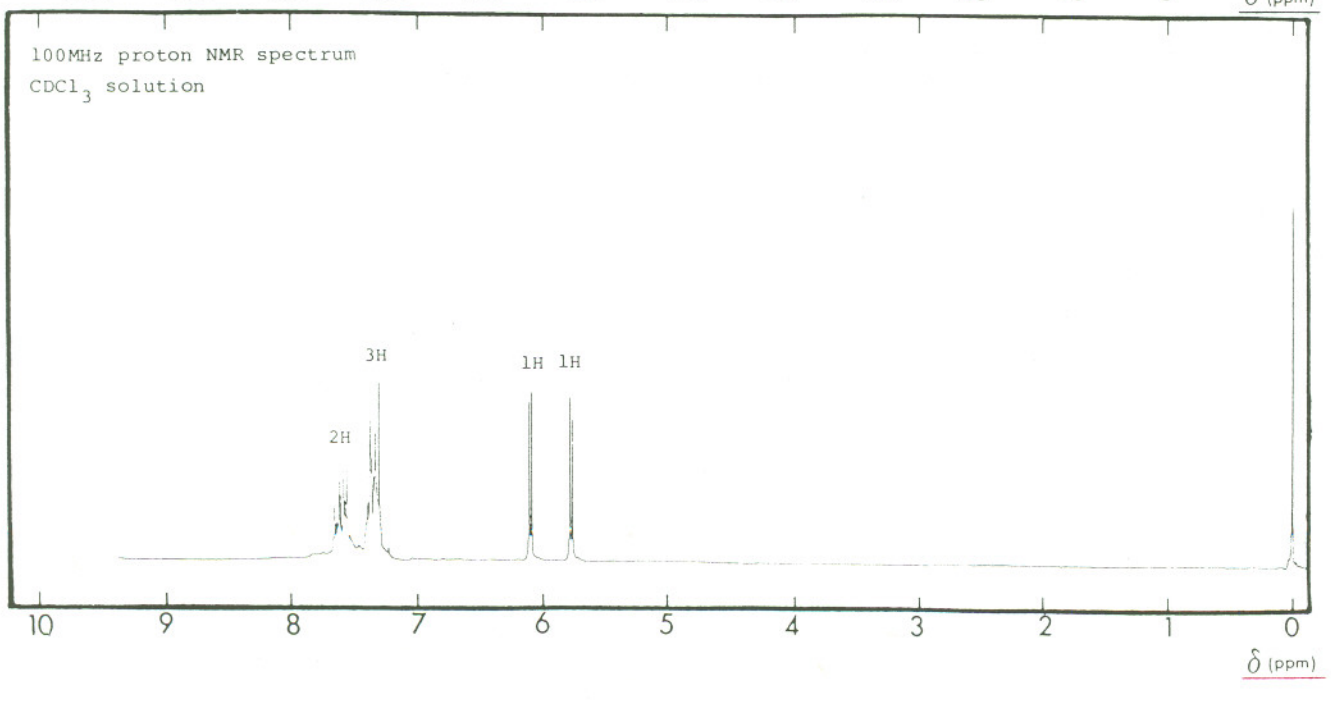
Problem 2


IR spectrum
(liquid film)
4000
3000
2000
1600
1200
800
ν (cm⁻¹)

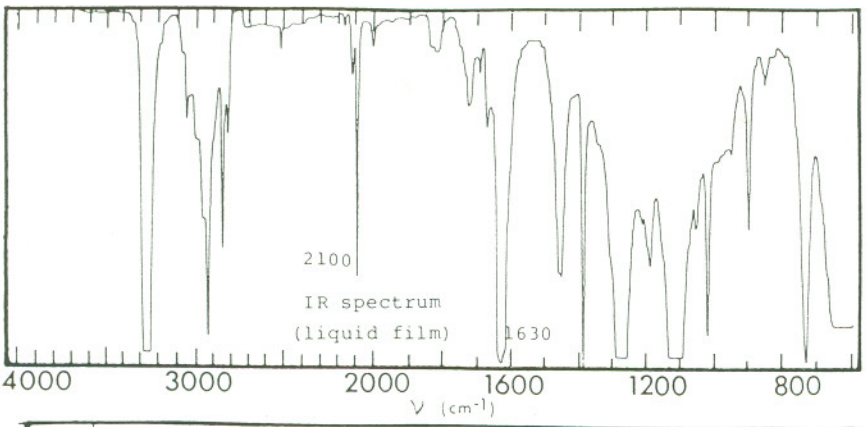

mass spectrum
% of base peak
90
70
50
30
10
103
77
M⁺ 182/184
20 40 60 80 100 120 140 160 180 200 220 240 260 280 300 320 340
m/e

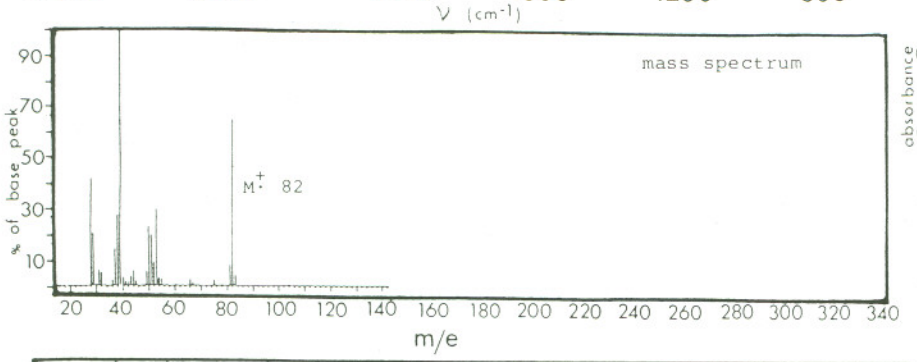

absorbance
0
0.3
0.6
0.9
1.2
1.5
UV spectrum
0.258mg/10ml EtOH
0.5cm cell
200 250 300 350
λ (nm)

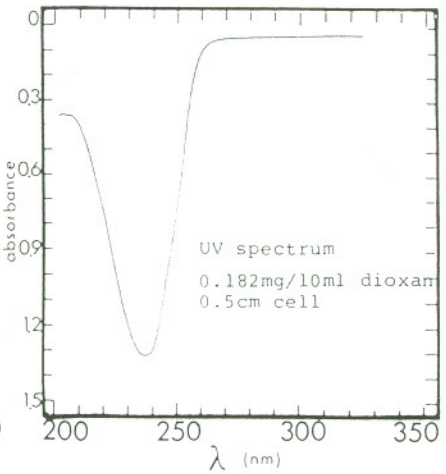

20MHz carbon-13
NMR spectrum
CDCl3 solution
*two peaks
overlapped
off-resonance decoupled
proton decoupled
220 200 180 160 140 120 100 80 60 40 20 0
δ (ppm)

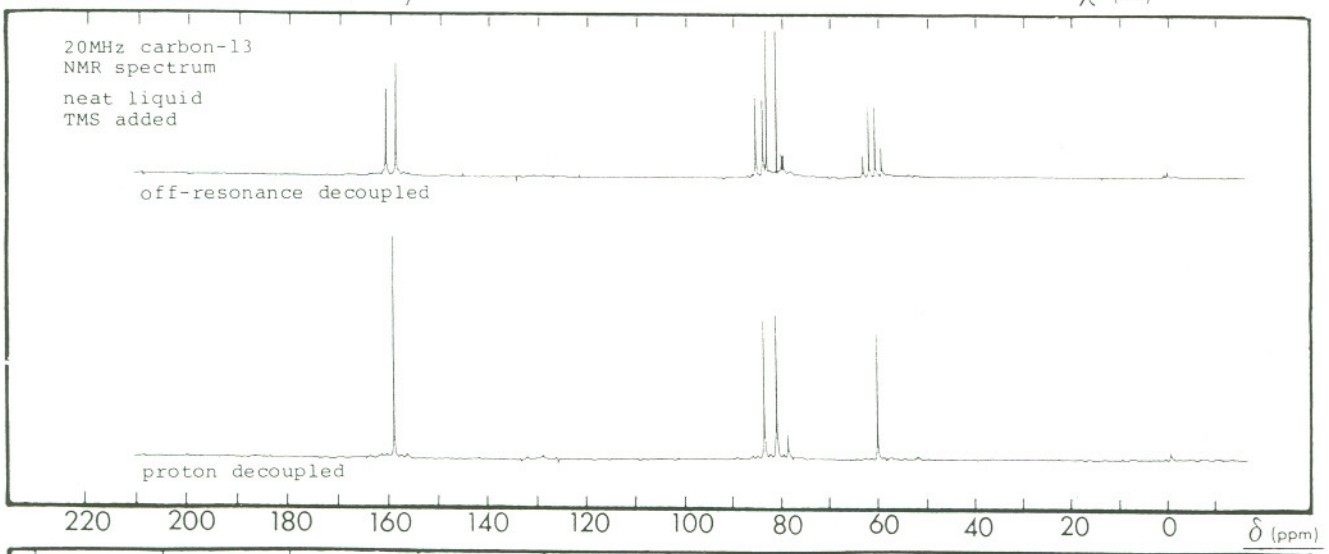

100MHz proton NMR spectrum
CDCl3 solution
3H
2H
1H 1H
10 9 8 7 6 5 4 3 2 1 0
δ (ppm)

Problem 3


2100
IR spectrum
(liquid film)
1630
4000
3000
2000
1600
1200
800
ν (cm⁻¹)


mass spectrum
% of base peak
M⁺· 82
m/e


absorbance
UV spectrum
0.182mg/10ml dioxan
0.5cm cell
λ (nm)


20MHz carbon-13
NMR spectrum
neat liquid
TMS added
off-resonance decoupled
proton decoupled
δ (ppm)

100MHz proton NMR spectrum
CDCl3 solution
50Hz
1
1
3
1
δ (ppm)